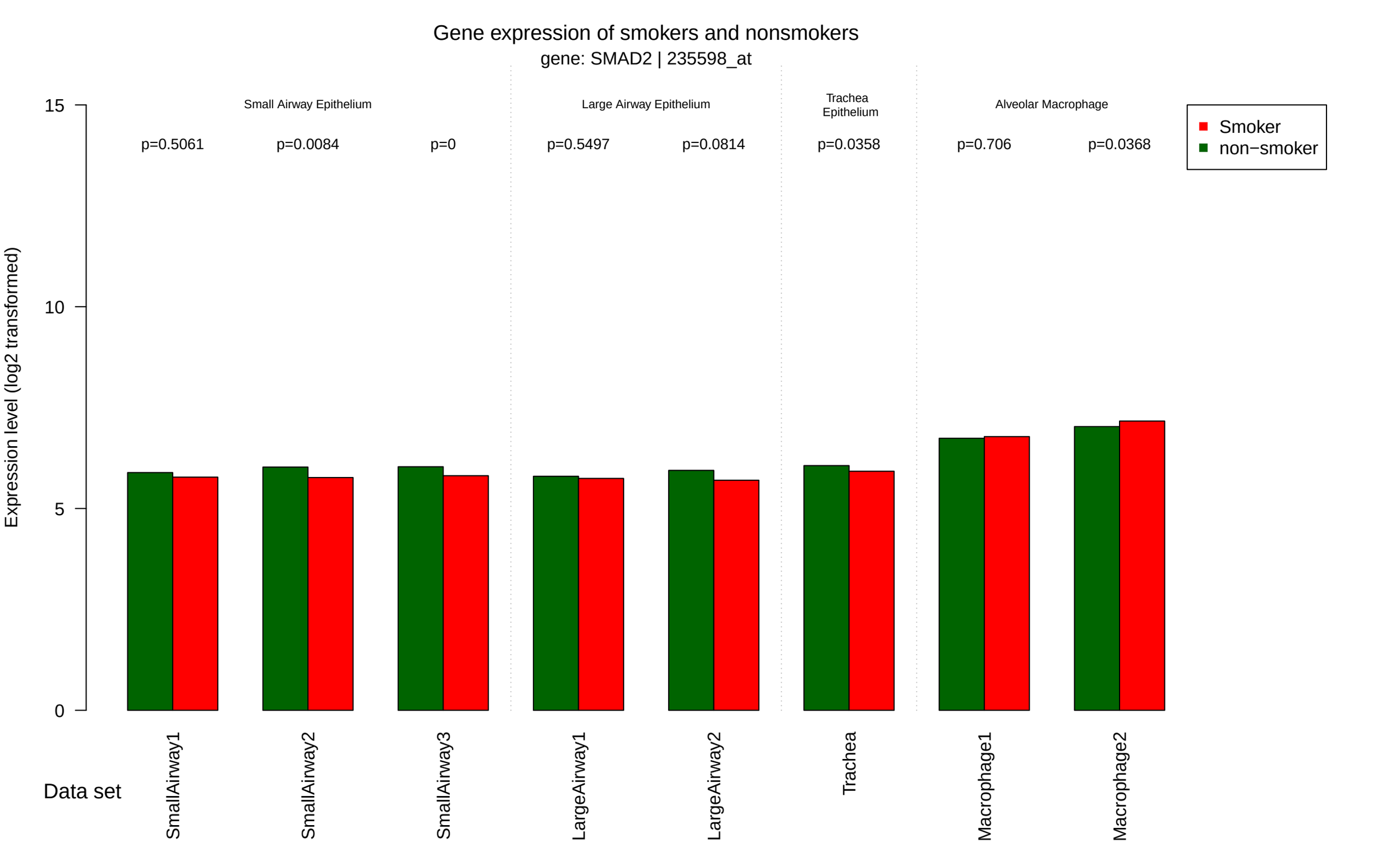
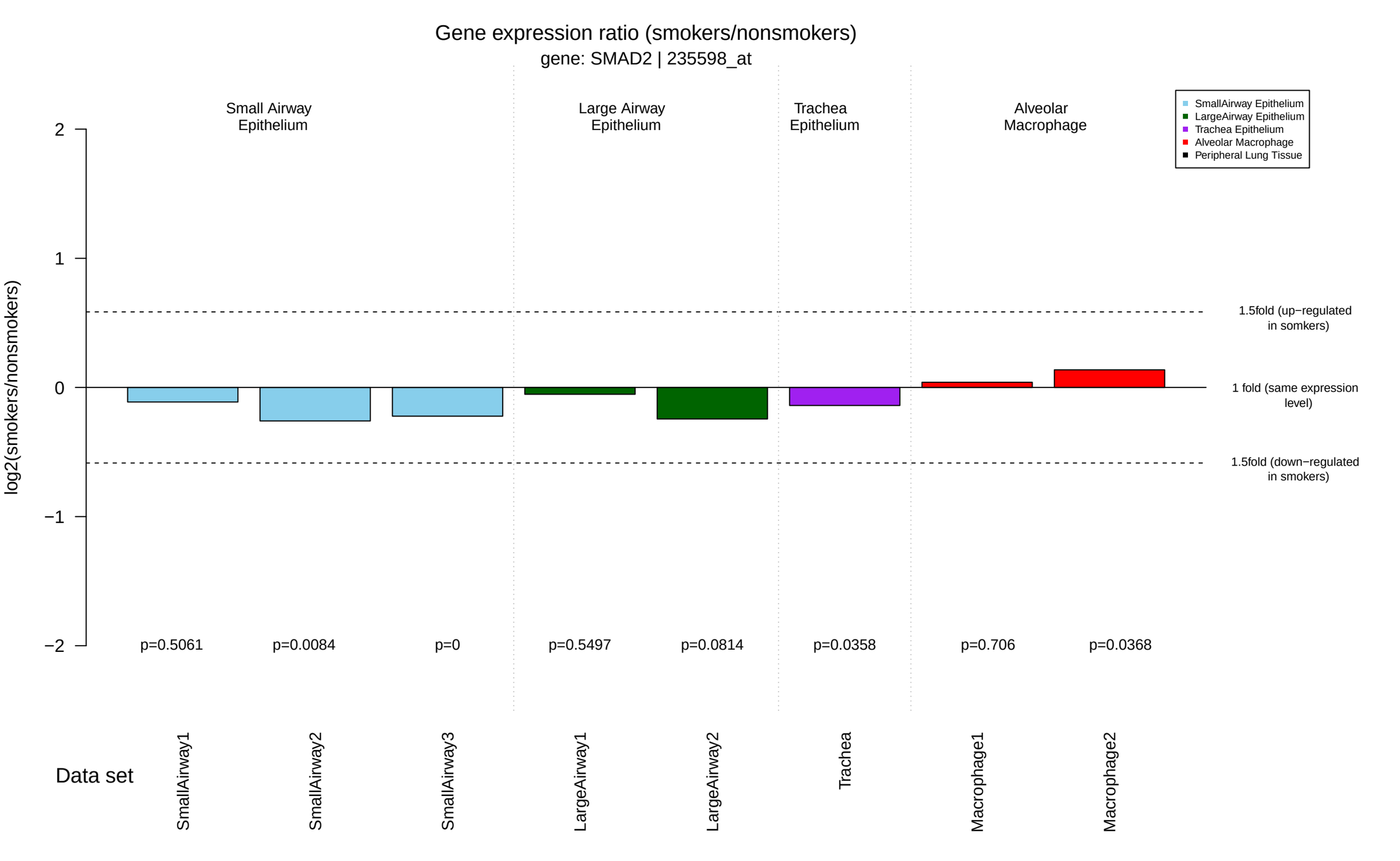
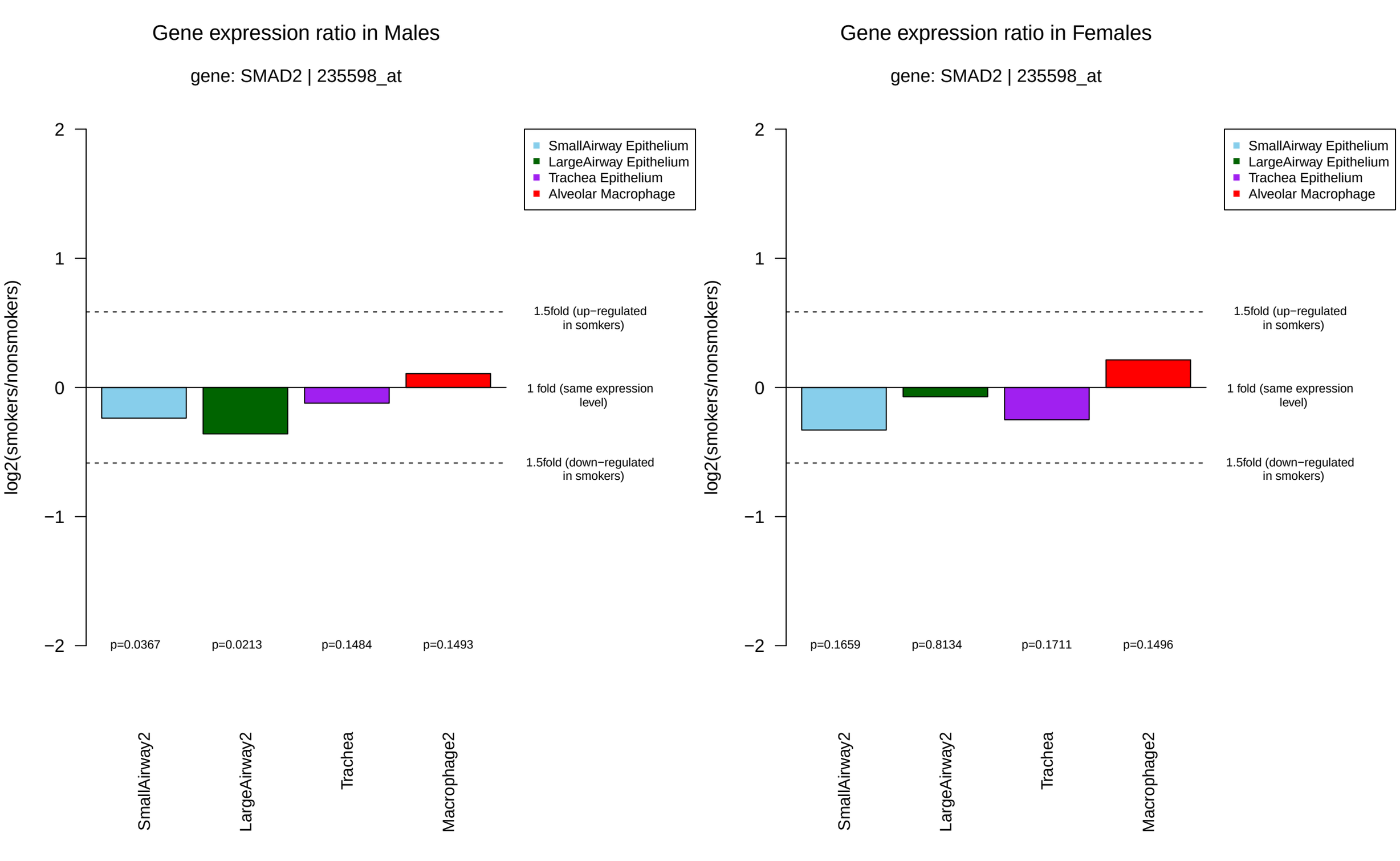
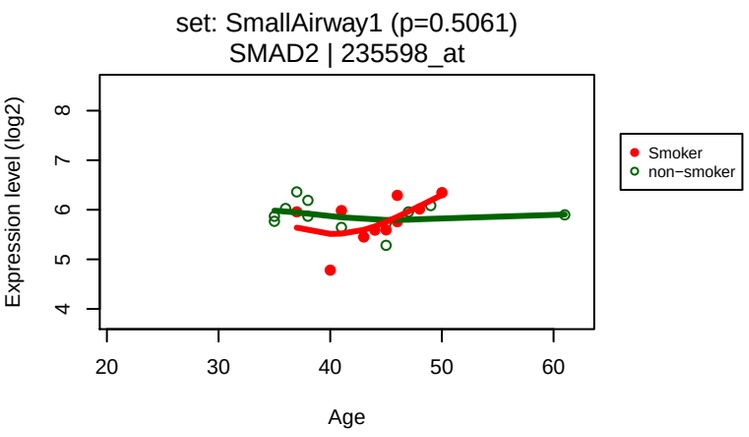
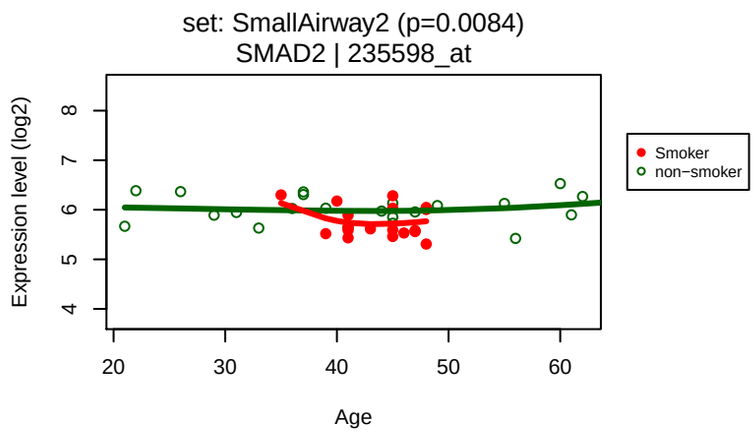
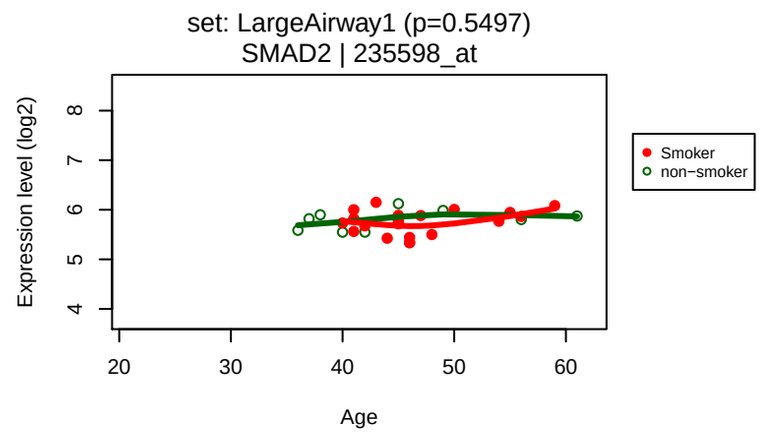

Gene expression of smokers and nonsmokers
gene: SMAD2 | 235598_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5061
p=0.0084
p=0
p=0.5497
p=0.0814
p=0.0358
p=0.706
p=0.0368
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

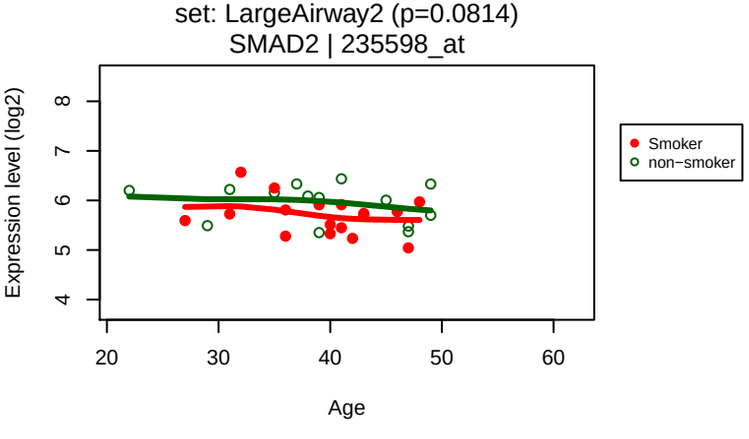

Gene expression ratio (smokers/nonsmokers)
gene: SMAD2 | 235598_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5061
p=0.0084
p=0
p=0.5497
p=0.0814
p=0.0358
p=0.706
p=0.0368
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

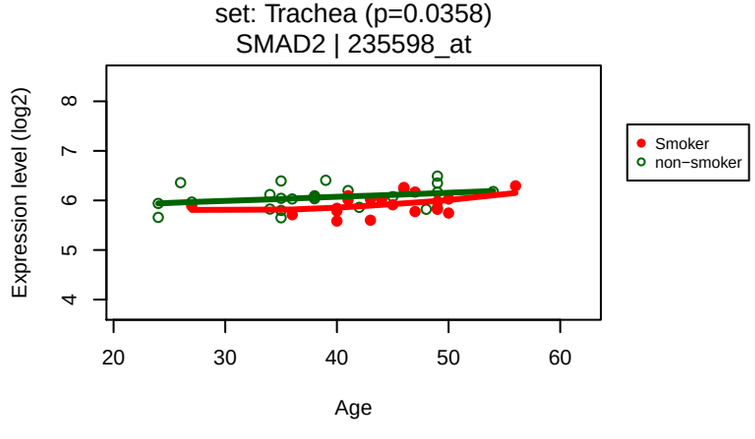

Gene expression ratio in Males
gene: SMAD2 | 235598_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0367
p=0.0213
p=0.1484
p=0.1493
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD2 | 235598_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1659
p=0.8134
p=0.1711
p=0.1496
SmallAirway2
LargeAirway2
Trachea
Macrophage2

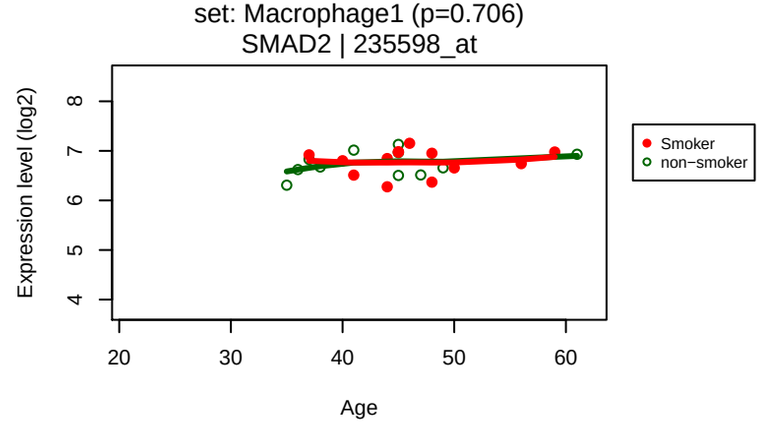

set: SmallAirway1 (p=0.5061)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker

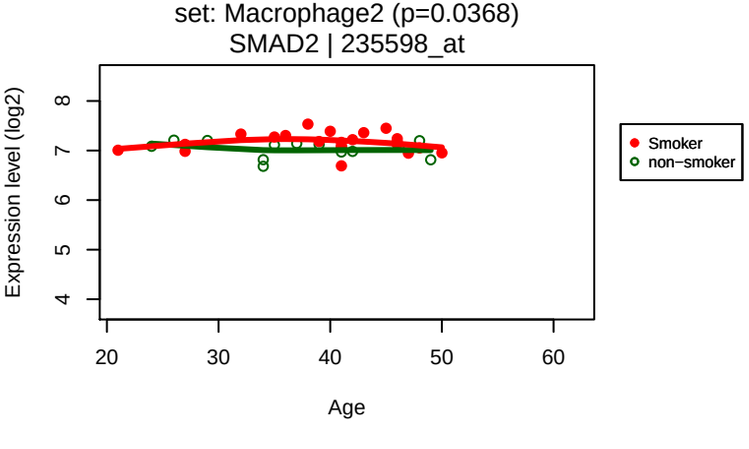

set: SmallAirway2 (p=0.0084)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker

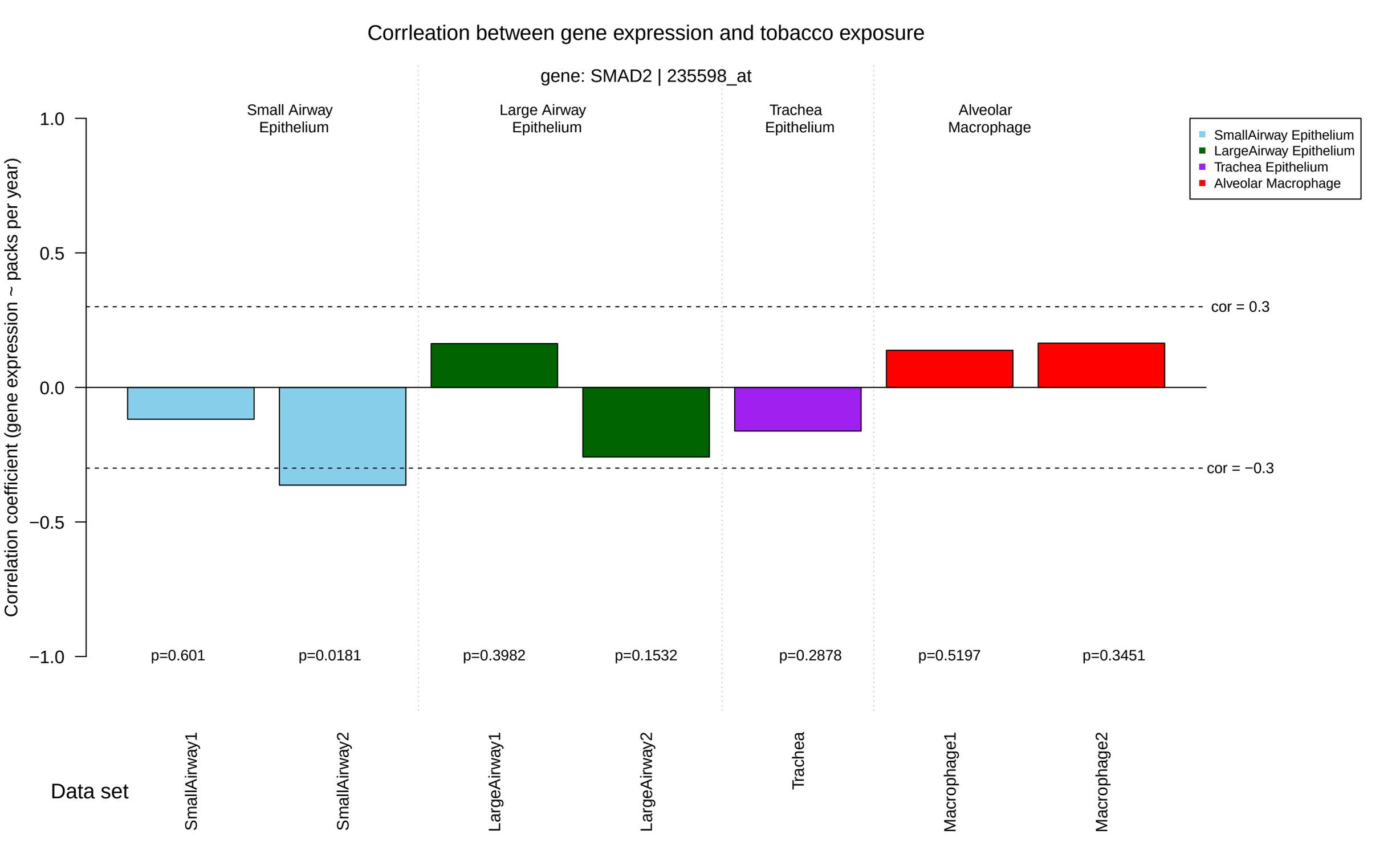

set: LargeAirway1 (p=0.5497)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0814)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0358)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.706)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0368)
SMAD2 | 235598_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD2 | 235598_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.601
p=0.0181
p=0.3982
p=0.1532
p=0.2878
p=0.5197
p=0.3451
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2